

Case 20: Point ignition and isotropic spread

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

Case ID: CASE20_PIG. This case verifies the coupled surface-fire and level-set rate response to a central point ignition in uniform fuel with zero wind and zero slope. Rotational symmetry supplies a diagnostic for directional bias or raster-orientation errors, while the formal decision tests the coupling between fire behavior and level-set normal velocity. The verification question is whether the required maximum spread rate is within 10 percent of 1.51 m min^{-1} ; isotropic arrival geometry is reported as supporting evidence but is not yet a separate acceptance component.

The scientific setup and historical expected response are reconstructed from Section 3.2 of the supplied *ELMFIRE Guide*. The Guide is used as reference provenance, not as an instruction or a substitute for evidence. Configuration statements below are checked against the generated GeoTIFFs and namelist; current results and decisions come only from case-local outputs and generated metrics. Where the generated experiment differs from the historical example, the report states that difference explicitly.

2 Mathematical formulation and numerical implementation

ELMFIRE advances the level-set field using $\partial\phi/\partial t + V_n|\nabla\phi| = 0$. With spatially uniform fuel, moisture, wind, and slope, $V_n = R_0$ is independent of azimuth. The zero contour must therefore evolve from the ignition neighborhood toward a circle. The final **vs** raster records the local normal spread rate in metres per minute because **SPREAD_RATE_IN_M** is enabled.

The surface-fire scalar supplied to the level set is the Rothermel no-wind/no-slope rate

$$R_0 = \frac{I_R \xi}{\rho_b \epsilon Q_{ig}}, \quad Q_{ig} = 250 + 1116 M_f, \quad (1)$$

where reaction intensity I_R , propagating-flux ratio ξ , bulk density ρ_b , effective heating number ϵ , and fuel moisture M_f are evaluated from the generated Fuel Model 3 and moisture inputs. This equation chain follows Andrews (2018), *The Rothermel Surface Fire Spread Model and Associated Developments*; the Guide supplies the configured known-response value.

3 Derivation of expected behavior

For constant R_0 , the front radius is $r(t) = r(0) + R_0 t$. The independently tabulated Fuel Model 3 response for the configured moisture and no-wind/no-slope condition is $R_0 = 1.51 \text{ m min}^{-1}$. Rotational symmetry is an invariant; the quantitative decision uses the rate because it directly tests the fire-behavior value passed to propagation.

The independent no-wind/no-slope reference is calculated from

$$R_0 = \frac{I_R \xi}{\rho_b \epsilon Q_{ig}}, \quad Q_{ig} = 250 + 1116 M_f,$$

using generated Fuel Model 3 and $(M_{1h}, M_{10h}, M_{100h}, M_{lh}, M_{lw}) = (0.06, 0.07, 0.08, 0.60, 0.90)$. The independent calculation gives $I_R = 549.174 \text{ kW m}^{-2}$ and $R_0 = 1.51183 \text{ m min}^{-1}$, reported to the Guide precision as $R_{\text{ref}} = 1.51 \text{ m min}^{-1}$ ($1.51183/0.3048 = 4.960 \text{ ft min}^{-1}$). The decision metric is therefore

$$e_R = \frac{|R_{\text{max}} - 1.51|}{1.51},$$

where R_{max} is the maximum finite final **vs** value, and the substituted acceptance test is $e_R \leq 0.10$.

4 Verification metrics and acceptance criteria

The head-ROS metric is the maximum finite value in the required final **vs** raster. It is selected because uniform forcing makes every mature front direction equivalent and the maximum avoids dilution by unburned nodata cells. A 10 percent relative tolerance follows the GUIDE comparison and allows raster and front-discretization effects while remaining sensitive to unit or coupling errors. The case passes only when the output exists and $|R_{\text{sim}} - 1.51|/1.51 \leq 0.10$.

5 Simulation configuration and predicted outcome

The case uses a 126×126 , 30 m grid in EPSG:32635, uniform Fuel Model 3, 6 percent one-hour moisture, flat terrain, zero wind, and a central ignition. The predicted output is an approximately circular arrival-time field with a uniform mature spread rate near 1.51 m min^{-1} .

Parameter	Configured value	Units	Role in the experiment
Grid	126×126 at 30	m per cell	Resolves the whole spatial response on the generated north-up raster
Projection	EPSG:32635	—	Fixes the map-coordinate and row-orientation convention
Time control	$\Delta t = 1$, $t_{\text{stop}} = 259200$	s	Whole-second step; stop time is an exact timestep multiple
Meteorology interval	3600	s	Defines weather-band timing and stalled-final compatibility
Live moisture	herb 60; woody 90	percent	Holds live-fuel damping fixed while the named mechanism is isolated
Evidence	prepared inputs; final case outputs	—	Separates configuration provenance from acceptance evidence

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

Whole-domain input configuration — data; WS=0 mph, WD=0 deg

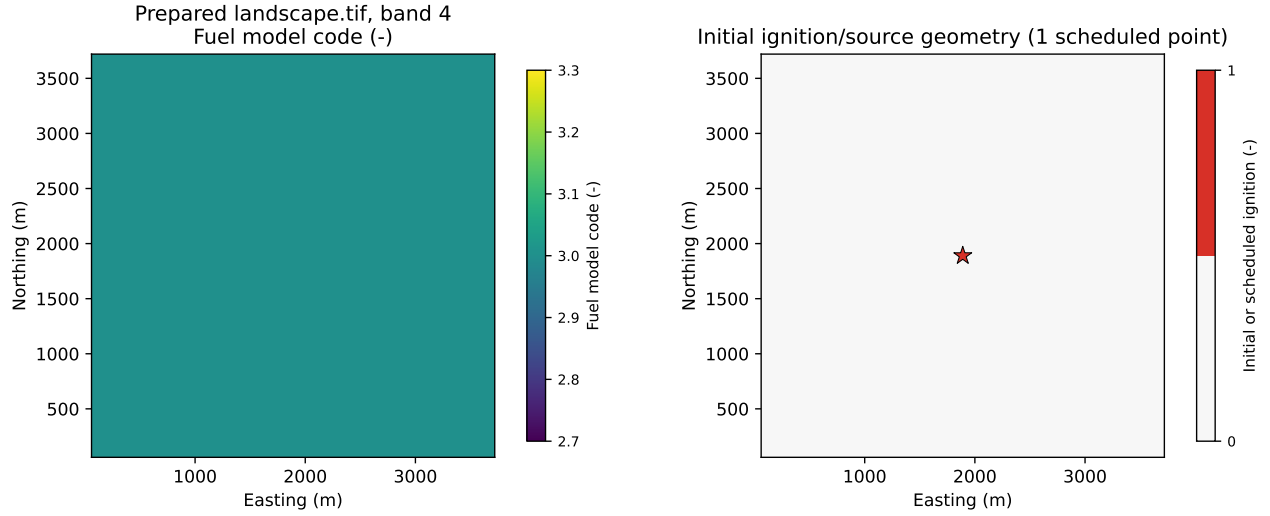


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The required evidence is generated only from case-local ELMFIRE outputs. The current postprocessing status is **PASS**; 1 of 1 required metric rows pass across 1 variant(s). A missing output is reported as NOT RUN or NOT EVALUABLE and is not interpreted as verification evidence.

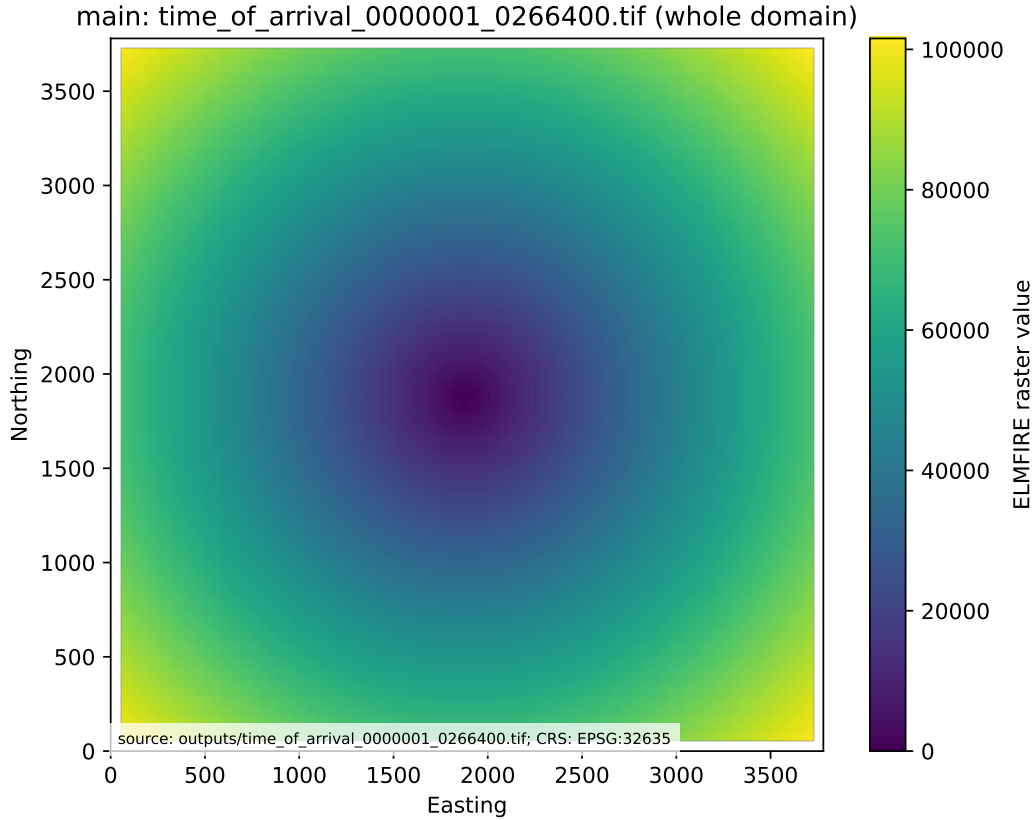


Figure 2: Final whole-domain ELMFIRE time of arrival for the displayed variant, in seconds. The figure annotation records the selected case-local GeoTIFF path and coordinate reference system; the generated metrics table, not visual appearance, determines acceptance.

7 Calculated metrics and verification decision

Variant / metric	Expected	Calculated	Status
main / head ROS	1.51 m/min	1.51183 m/min	PASS

The displayed overall decision is **PASS**. When `outputs/metrics.json` exists, `scripts/postprocess.py` generates every table value and status; when it is absent, the standalone wrapper renders NOT EVALUABLE with no invented rows.

8 Technical interpretation

A rate failure indicates a fire-behavior, units, or output-selection problem. Strong directional distortion with an acceptable maximum is a diagnostic warning and must not be claimed as verified until a shape metric and tolerance are added. No scientific conclusion should be drawn when the status is NOT RUN or NOT EVALUABLE.

9 Scope and limitations

This is deterministic code verification for one fuel and moisture state. It does not validate Fuel Model 3 against experimental fires or quantify symmetry with a dedicated shape metric. This case provides implementation verification evidence only.